

Soumyadip Sil

B.Tech Student in Electronics and Communication

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Summary

B.Tech (ECE) graduate with hands-on experience building and deploying full-stack and agentic AI systems, including a production MCP-based agent with LLM confidence-scoring for hallucination prevention. Proficient in Python, TypeScript, SQL, and Node.js/Express, with practical exposure to the Model Context Protocol (MCP), LLM orchestration, and cloud deployment (Render, Vercel). Strong foundation in C/C++ and systems programming, with a track record of shipping working, deployed projects rather than just prototypes.

Projects

1. Autonomous Multi-Task Assistant Robot | C++, Arduino IDE

- Built an Arduino-based assistant robot integrated with the Google Gemini API to process voice commands, answer queries, execute tasks in real-time, and take both voice and manual mobility commands.
- Programmed autonomous navigation logic using ultrasonic sensors to detect physical obstacles and automatically adjust the robot's path without user input.
- Linked the physical hardware to a mobile application interface to allow for both remote manual control and live video monitoring.

2. Context Protocol Agent | Node.js, Express, MCP

- Built a full-stack AI agent that orchestrates a Notion MCP server to securely query and retrieve private workspace data via natural language.
- Designed a ReAct-style orchestration loop (search → retrieve → synthesize) with an Express.js backend and a minimal, zero-layout-shift frontend.
- Engineered a strict JSON confidence-scoring layer using NVIDIA Nemotron to cross-reference retrieved documents, flagging conflicting or insufficient sources instead of letting the model hallucinate an answer.
- Deployed as a persistent Node.js web service on Render, with the agent-to-tool interface decoupled so the same orchestration logic could extend to other MCP-compatible tools (Slack, GitHub, Postgres).

3. Blog website - Afterhours | Next.js, TypeScript, Supabase, Tailwind CSS

- Built and deployed a full-stack personal blog application using Next.js and TypeScript, styled with Tailwind CSS with Supabase as the backend database.
- Configured continuous deployment through Vercel, connecting the GitHub repository for automatic build and release on every push.

Skills

Languages: JavaScript/TypeScript, Python, Java, C/C++, SQL, HTML/CSS
AI & Architecture: AI Agents, LLM Integration(OpenRouter/NVIDIA NIM), Model Context Protocol (MCP)
Tools & Frameworks: Node.js, Express.js, REST APIs, Git, GitHub, Linux

Education

Techno Main Salt Lake(MAKAUT)	Kolkata
B.Tech in Electronics and Communication Engineering	08/2022 - 08/2026
Madhyamgram High School	04/2020 - 05/2022
Class XII (WBCHSE) 86.2%	
Madhyamgram High School	04/2019 - 03/2020
Class X (WBBSE) 87%	

Languages

English Proficient	Hindi proficient	Bengali Proficient
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